

Comparing Colour Thresholding and a Vision Language Model for Building Matching in
Satellite Images

An Exploratory Study Using OpenStreetMap References

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github.com/erenesinler23/geoai-feature-detection

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Abstract

I developed a pipeline to explore red roof detection using a colour rule, OWL ViT and a hybrid that combines their outputs. I compared their predictions with OpenStreetMap building footprints in seven Sentinel 2 urban crops. At an intersection over union threshold of 0.25, the colour rule achieved precision between 4.84% and 24.94% and recall between 0.19% and 7.32%. OWL ViT and the hybrid produced no matched buildings. The reference data includes buildings of every roof colour, leaving roof colour accuracy unmeasured; the scores describe building overlap. To investigate the empty outputs, I repeated inference on three recent live crops, checked coordinate conversion and tested the model on a photograph and three external overhead images. It located both birds in the photograph and matched buildings in two overhead images. Those building matches disappeared when I deliberately degraded the images, suggesting that image detail contributed to the results. Other causes remain possible. The project provides a working application and archived experiments that document how the methods behaved and where they failed.

1. Introduction

During my internship at BÎTES, an ASELSAN subsidiary (BÎTES, no date), I used time allocated for independent study to propose and develop this project. My initial work involved learning about geospatial data formats and exploring the team's two long term goals: photo geolocation and natural language queries on satellite imagery. From this starting point, I developed the study presented in this paper.

The NATO Communications and Information Agency document GeoAI Solutions to Support Selected NATO Use Cases, dated 12 May 2026, provided further motivation through its interest in extracting building footprints (NCIA, 2026; Ministry of Development and Technology, 2026). It is a request for information issued for market research and planning. I used it to understand the wider context for geospatial feature extraction. I undertook this study independently of any assigned BÎTES deliverable or NATO commission. It does not assess compliance with the agency's requirements. I use roof colour as a visual cue, leaving construction material unverified.

The project began with a practical question: could a simple colour rule and a model that accepts text prompts help locate red roofs in satellite imagery? The available reference data lets

me assess a narrower question: how closely do their candidate regions match mapped building footprints? OSM supports a `roof:colour` tag, and the parser retains it when present (OpenStreetMap contributors, no date). However, the experiment does not have independently checked roof colour labels. The reference set therefore includes buildings of every colour. A spatial match supports building overlap, while the colour of that roof remains unverified.

I built the colour detector, integrated OWL ViT and added a hybrid that uses model boxes to filter colour candidates. A shared pipeline handles imagery, coordinates and evaluation, while a web application lets users search for places and inspect the results. The paper retains red roof detection as its motivation and reports the limits of the available evaluation. Photo geolocation remains outside its scope. The application in Appendix A accepts searches beyond the seven experiment locations.

2. Related Work

Minderer et al. (2022) introduced OWL ViT, an object detector that accepts text queries. It transfers representations learned from images and text to object detection. CLIP uses contrastive learning to connect images with natural language descriptions (Radford et al., 2021). These approaches make it possible to request a category without training a separate detector for every label. Their performance on small objects in satellite imagery still needs to be tested.

GeoChat is a vision language model designed for remote sensing (Kuckreja et al., 2024). Its treatment of small objects and image scale makes it relevant to this study. I discuss it as related work but did not test it. Comparing it with this pipeline would require attention to the models' different architectures, training data and image preparation.

Sentinel 2 provides optical imagery with visible bands at a spatial resolution of 10 metres (Drusch et al., 2012). The pipeline finds imagery through a SpatioTemporal Asset Catalog, or STAC, which describes geospatial assets and their locations (SpatioTemporal Asset Catalog, no date). I use the catalogue visual asset as an RGB image. Its values represent display colours rather than calibrated reflectance.

3. System Design and Implementation

3.1 Geospatial Data Formats

The project combines raster images with vector building footprints. The raster coordinate reference system and affine transform connect each pixel to a position on the map. Reading a window from a Cloud Optimised GeoTIFF avoids downloading the whole image. OSM supplies building polygons, and the experiment saves them as GeoJSON alongside the crop.

3.2 Initial Implementation

The original implementation used separate scripts for fetching imagery, extracting candidates and comparing footprints. As the project grew, I moved the shared work into a package. The current version keeps imagery retrieval, map parsing, detection and evaluation separate. The command line runner and Flask application use the same functions, which makes changes easier to test.

3.3 A Shared Detector Interface

Each detector implements the same interface. It takes three image channels, a transform and a coordinate reference system, then returns geometries with unique detection identifiers. The evaluator can compare these outputs without needing to know how each detector produced them.

```
class Detector(ABC):
    @abstractmethod
    def detect(self, r, g, b, transform, raster_crs):
        raise NotImplementedError

    @property
    @abstractmethod
    def name(self):
        raise NotImplementedError
```

3.4 Validation Procedure and Its Metrics

The revised evaluation first defines the usable part of the image. It rejects incomplete crops and uses the Sentinel scene classification layer to exclude missing pixels, cloud shadows, clouds and snow. Codes 4, 5, 6 and 7 are retained. A crop is accepted only when no more than 10% of its pixels are excluded. The OSM request covers the bounds of this usable area. Only complete building footprints inside it enter the reference set. Predictions are clipped to the usable area. This makes the spatial policy explicit, although a prediction of an excluded partial building can still be unmatched.

Matching is performed in the raster coordinate system, measured in metres. Intersection over union, or IoU, is the intersection area divided by the union area of a prediction and a reference polygon. A valid pair needs IoU of at least 0.25. Each prediction can match one reference and each reference can match one prediction. The assignment first maximises the number of valid pairs, then their total IoU. Duplicate predictions therefore cannot earn repeated credit for one building. Sensitivity is checked at IoU thresholds of 0.10 and 0.50.

Before matching, the evaluator removes regions wider or taller than 500 metres. I chose this permissive limit to remove very large regions; its suitability as an individual building size limit remains unverified. The number removed is saved for each method. Precision is the matched share of retained predictions. Recall is the matched share of eligible reference buildings. False discovery rate is the unmatched share of retained predictions. These scores are conditional on the size filter and on the OSM reference set.

```
precision = 100 * matches / predictions if predictions else None
```

```
recall = 100 * matches / references if references else None
```

```
f1 = 200 * matches / (predictions + references) if predictions + references else None
```

The interface displays N/A when a denominator is empty. With reference buildings but no predictions, recall and F1 are zero, while precision and false discovery rate are undefined. This avoids presenting an empty detector as having a perfect error score. Tests cover duplicate predictions, boundary contact, empty inputs, coordinate conversion and the size filter.

3.5 The Hybrid Detector

The hybrid keeps a colour candidate when its centroid lies inside a model box that passes the size filter. Both checks use a projected coordinate system. This is a simple way to filter colour candidates using the model output, with no new detection architecture. It also makes the hybrid dependent on the model: if no model boxes survive, it returns no candidates.

```
hybrid = fuse_detections(naive, owlvit, max_size_m=500)
result = validator.validate(hybrid, buildings, "hybrid", raster_crs)
```

The live application uses the same matching procedure. Appendix A shows its current interface with examples from London, Prague and Dubrovnik. Those examples come from separate live searches and sit outside the seven location experiment.

4. Experimental Setup

The corrected batch covers Ankara, Marrakech, Nottingham, London, Istanbul, Prague and New Delhi. Each location contributes one crop. All these cities had already appeared in exploratory work, so this is not an independent test set. Exact crop centres are fixed in benchmark.json. The same colour and confidence thresholds are used throughout this batch, without adjustments after viewing its results.

The imagery search covers 1 April to 31 August 2026 and requires reported scene cloud cover below 40%. Candidate scenes are ordered by cloud cover and identifier. The fetcher takes the first complete crop that passes its local quality check. Each crop is 300 by 300 pixels, with a nominal width and height of 3 kilometres. Table 1 identifies the saved scenes and eligible reference counts. The manifests contain acquisition times, exact transforms and the OSM snapshot timestamps.

Table 1. Scenes and eligible OSM building counts for the corrected experiment. Full metadata is saved with each crop.

Location	Sentinel 2 scene identifier	Reference buildings	Excluded at boundary or mask
Ankara	S2A_MSIL2A_20260701T084501_R064_T36TVK_20260701T134111	6769	185
Marrakech	S2B_MSIL2A_20260729T110619_R137_T29SNR_20260729T134723	13402	154
Nottingham	S2C_MSIL2A_20260525T110621_R137_T30UXD_20260525T144513	8325	725
London	S2C_MSIL2A_20260525T110621_R137_T30UXC_20260525T144513	8801	1978
Istanbul	S2C_MSIL2A_20260426T085601_R007_T35TPF_20260426T143618	5819	242
Prague	S2A_MSIL2A_20260814T100041_R122_T33UVR_20260814T151116	8237	998
New Delhi	S2B_MSIL2A_20260617T052649_R105_T43RGM_20260617T104156	1435	119

The colour rule requires red above 120, red minus green above 25, and red minus blue above 25. It groups connected pixels and returns a box around each region. I kept these values from the earlier exploratory work and applied them to every location in the corrected batch, including Nottingham, which had previously used different settings. No further tuning took place during the batch, although the values still reflect choices made during exploration.

The exact model prompts are “a reddish or ochre-colored building” and “a red or terracotta rooftop”. Its confidence threshold is 0.01. Model and processor files use google/owlvit-base-patch32 at revision cbc355fb364588351c5d51c7f74465e8e7ec6f72, and inference uses evaluation mode. The archive records the actual package versions, all raw model scores and boxes, and the settings used. No duplicate suppression is applied, so repeated boxes are penalised during matching.

The corrected London crop uses a centre of longitude 0.1276 degrees west and latitude 51.5072 degrees north. Its centre is now explicit rather than dependent on an unrecorded geocoding response. The original and corrected experiments also differ in matching, reference handling and some settings. Their percentages cannot be compared as if matching were the only change.

The revised OSM parser includes building ways and multipolygon relations, preserves available roof colour tags and repairs invalid polygon geometry before use. The reference set is saved rather than fetched again during replay. It remains an imperfect map reference: a missing building can make a valid prediction appear unmatched, and a current footprint may differ from the building visible on the image acquisition date.

4.1 Diagnostic Tests

After inspecting the original results, I added diagnostic tests on 21 September 2026. These were exploratory checks, separate from the fixed experiment in Tables 1 to 3. I audited 20 saved run folders, including repeated scenes, and repeated inference on the recent London, Prague and Dubrovnik crops identified in Appendix A. The folders do not represent 20 independent scenes. These examples have now been used to investigate the system and cannot serve as an untouched test set.

The checks used the recorded OWL ViT checkpoint, Transformers 5.17.0 and CPU execution. A known pixel box was converted through a 10 metre affine transform and checked in projected coordinates. I also tested a photograph containing two birds with the prompt “a photo of a bird” (Narsil, no date). The photograph provides a simple check of whether the inference path can locate obvious objects. Roof detection and general accuracy require separate evaluation.

For the three recent satellite crops, I compared the original two prompts with “a satellite image of a building”. Confidence remained at 0.01, the maximum box dimension at 500 metres and matching IoU at 0.25. Predictions were clipped to the same usable area and evaluated against the same saved footprints. This check asks whether a broader building prompt changes the output. Since it drops the roof colour requirement, it addresses a different target from the original experiment.

I used all three images in the external satellite building segmentation dataset’s small validation archive, together with its supplied building box annotations (keremberke, no date). The images are 500 by 500 pixels. Each was reduced to 25 by 25 with area averaging, then enlarged to its original dimensions with bilinear interpolation. The generic building prompt and confidence threshold of 0.01 were unchanged. Matching used unique pairs at IoU 0.50 against annotation boxes. No geographic size filter was applied because these inputs had no verified ground sampling distance in this test. Duplicate suppression was not added. The different reference geometry and matching threshold mean these scores should not be ranked directly against the OSM experiment.

5. Results

5.1 The Naive Colour Rule

Table 2 reports results after the size filter at IoU 0.25. Precision, recall and F1 are percentages. The reference count for each crop is listed in Table 1. The measurements come from the corrected run archives. The historical figures below show the earlier exploratory outputs.

Table 2. Colour rule building matching results after the size filter at IoU 0.25. Precision, recall and F1 are percentages.

Location	Precision %	Recall %	F1 %
Ankara	4.84 133 of 2750	1.96 133 of 6769	2.79
Marrakech	7.78 26 of 334	0.19 26 of 13402	0.38
Nottingham	7.40 37 of 500	0.44 37 of 8325	0.84
London	7.35 82 of 1116	0.93 82 of 8801	1.65
Istanbul	23.77 367 of 1544	6.31 367 of 5819	9.97
Prague	24.94 481 of 1929	5.84 481 of 8237	9.46

Location	Precision %	Recall %	F1 %
New Delhi	5.05 105 of 2081	7.32 105 of 1435	5.97

The colour rule achieved precision from 4.84% to 24.94% and recall from 0.19% to 7.32%, indicating limited agreement with individual OSM footprints. Several factors could account for the variation between locations, including image appearance, building size, map completeness and how many buildings meet the colour rule. Their individual effects remain unresolved in this experiment. As defined in Section 4, the reference data also leaves roof colour unverified.

Real satellite scene, S2B_MSIL2A_20260729T110619_R137_T295NR_20260729T134723



Figure 1. Original Marrakech satellite image retained from the exploratory version.

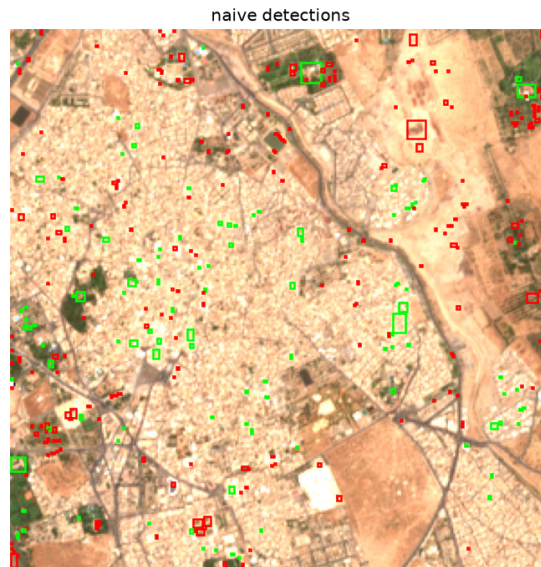


Figure 2. Original Marrakech colour output. The green markers show the earlier intersection rule, not the corrected IoU matches in Table 2.

Real satellite scene, S2C_MSIL2A_20260525T110621_R137_T30UXD_20260525T144513



Figure 3. Original Nottingham satellite image retained from the exploratory version.

naive detections



Figure 4. Original Nottingham colour output. Its settings and marker meanings belong to the earlier experiment.

5.2 OWL ViT and the Hybrid Detector

Table 3 reports model candidates passing the confidence threshold, model predictions retained for evaluation, and retained hybrid predictions. Counts before filtering are included because a zero final result can have different causes. The complete archives include scores and geometries for inspection.

Table 3. OWL ViT candidate counts, retained predictions and building matches, with retained hybrid predictions and matches.

Location	OWL ViT counts	Hybrid counts
Ankara	2 above confidence 0 retained 0 matched	0 retained 0 matched
Marrakech	3 above confidence 0 retained 0 matched	0 retained 0 matched
Nottingham	0 above confidence 0 retained 0 matched	0 retained 0 matched
London	1 above confidence 1 retained 0 matched	3 retained 0 matched
Istanbul	0 above confidence 0 retained 0 matched	0 retained 0 matched
Prague	1 above confidence 0 retained 0 matched	0 retained 0 matched
New Delhi	0 above confidence 0 retained 0 matched	0 retained 0 matched

Only London retained a model prediction in the corrected batch. That prediction did not match an eligible building at IoU 0.25. The hybrid retained three London candidates, none of which matched. The other six crops had no retained model or hybrid predictions. Precision is therefore zero for the nonempty London outputs and undefined for the empty outputs. Recall and F1 are zero for both methods across the seven crops. These results describe this checkpoint, its prompts and the stated filters.

5.3 Photograph Diagnostic

The original diagnostic applied the model to a photograph of rooftops in Prague. It used image coordinates because the photograph has no geographic transform. A box was considered size plausible when its width and height were each below 25% of the image dimensions. This relative image size rule has no direct equivalence to the physical size limit used for satellite crops.

Control test, 93/124 plausible detections (green = plausible, red = oversized)

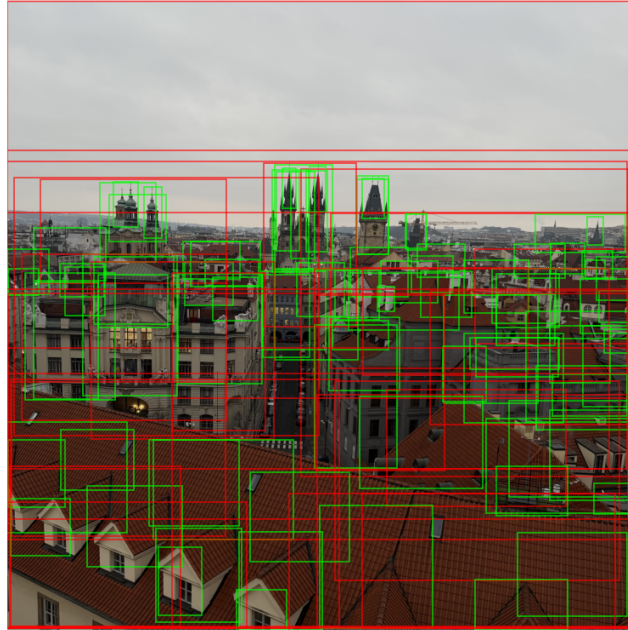


Figure 5. Original photograph diagnostic. Green boxes pass its relative size rule; they are not verified correct detections.

A repeat of the photograph diagnostic using the recorded model revision and a confidence threshold of 0.01 reproduced the original counts: 124 candidates passed the confidence threshold and 93 passed the relative size rule. The new run saves its boxes, scores and settings. The 93 retained candidates have no manual labels confirming which are roofs. Viewpoint, scale and scene content also differ from the satellite crops, leaving the effect of resolution mixed with those other changes.

5.4 Additional Application Examples

Figure 6 preserves the earlier Istanbul colour output. Its dense boxes show how many candidates the earlier detector generated; their accuracy remains unverified. The current webpage

examples appear in Appendix A, with their locations and saved run identifiers. The main numerical results in Tables 1 to 3 come from the fixed experiment archives.



Figure 6. Original Istanbul colour output. This figure illustrates the application and is not a plot of the corrected matching results.

5.5 Additional Diagnostic Results

The coordinate check agreed with the expected projected box within 0.001 metres, and all 18 existing unit tests passed. Their coverage is limited to the operations tested. On the bird photograph, two predictions exceeded 0.10 confidence, with scores of 0.598 and 0.412. Visual inspection showed a box around each bird. This result confirmed that the inference path could locate obvious objects in at least one image, something the unlabelled Prague rooftop photograph had left uncertain.

The original prompts again left no retained predictions on the three recent satellite crops. London produced one candidate above 0.01, with a highest score of 0.01326; its box measured about 2986 by 2970 metres. Prague also produced one candidate, scoring 0.01109, with a box about 2977 by 2986 metres. Both exceeded the 500 metre dimension limit. Dubrovnik's highest score was 0.00773, so it produced no candidate above the confidence threshold. Removing the size limit would expose the large boxes but would not make them useful building predictions.

With the generic prompt, London and Prague gained a few OSM matches (Table 4). Each precision value came from only a handful of predictions, and recall stayed very low. Most reference buildings still had no match.

Table 4. Generic building prompt on the recent live crops. Precision and recall are percentages.

Scene	Retained	Matches	Precision	Recall
London	3	1	33.33	0.0113
Prague	4	2	50.00	0.0349
Dubrovnik	1	0	0.00	0.0000

The external samples are identified by their supplied filenames. The model matched eight of nine annotated buildings in one Tripoli image and four of six in the other, while the Mykolaiv image had no predictions above 0.01. All predictions in these images had confidence below 0.05. The results therefore depend strongly on a permissive threshold, and duplicate or loose boxes still reduce precision.

Table 5. External building controls at IoU 0.50. Counts use the original images; the final column gives matches after degradation.

Image	Labels	Boxes	Matches	Degraded matches
Mykolaiv R3C3	5	0	0	0
Tripoli R3C1	9	27	8	0
Tripoli R3C3	6	8	4	0

Precision and recall were 29.63% and 88.89% for Tripoli R3C1, and 50.00% and 66.67% for Tripoli R3C3. After degradation, the first produced no predictions and the second produced one unmatched prediction. Mykolaiv produced none in either condition. Reducing image detail removed every building match in the two Tripoli samples. Section 6 considers how far this observation can help explain the Sentinel results.

6. Discussion and Limitations

The project combines a red roof colour rule, a text prompted detector and a simple hybrid in a working search application. Its main contribution is the implementation and the archived

evidence, including the unsuccessful results. Investigating the empty outputs proved useful: in some runs nothing reached the confidence threshold, while in others the size filter removed the model boxes. The application can therefore return an empty result even where buildings are visible.

The main evaluation problem is the reference data. I started with red roof detection, but the available footprints describe buildings of every colour. This mismatch can lower recall, and an overlapping footprint cannot confirm that a predicted roof is red. A proper red roof evaluation needs independently checked labels, a clear definition of the target colour and rules for ambiguous roofs. It should also record who checked the labels.

Image detail is a practical limit in these scenes. The median mapped footprint occupies an area equivalent to 1.28 source pixels in the recent London crop, 2.73 in Prague and 1.32 in Dubrovnik. These area equivalents come from dividing footprint area by the 100 square metre pixel area. They leave the proportion of roof within each pixel unknown. Alignment and mixed pixels make roof appearance harder to interpret. Enlarging an image cannot restore details that were absent from the source.

The degradation test in Section 5.5 kept scene content fixed, giving a more direct comparison than the original Prague photograph. Both external images with building matches lost those matches after degradation. With only three samples, however, the test is too small to establish a general relationship between resolution and accuracy. A simulation of the Sentinel sensor would require a different procedure from the resizing used here. The original ground sampling distances also remain unknown. Changing the prompt also affected the live results, although recall with the generic building prompt stayed below 0.04%. Model training, image preparation and object scale could all contribute to the poor performance.

The revised matching rule addresses repeated credit for loose or duplicate predictions. It still depends on choices about IoU, size limits and border objects. I used IoU 0.25 for this exploratory comparison; an appropriate threshold for other tasks would need its own justification. The saved threshold sensitivity results make its effect visible. OSM polygon accuracy and image alignment can also affect IoU, especially for small buildings. For the colour rule, precision at IoU 0.10, 0.25 and 0.50 was Ankara 13.56%, 4.84% and 0.76%; Marrakech 17.96%, 7.78% and 0.30%; Nottingham 15.80%, 7.40% and 0.80%; London 21.68%, 7.35% and 0.54%; Istanbul 55.25%,

23.77% and 2.40%; Prague 48.16%, 24.94% and 3.21%; New Delhi 15.67%, 5.05% and 0.72%. The full sensitivity results are included with the saved runs.

The experiment contains one selected scene from each of seven cities. I had already explored these locations when I chose the shared parameters, so the batch cannot tell us how well the system works on unseen cities. It also offers too little independent evidence for a reliable confidence interval for generalisation. A larger evaluation should separate tuning locations from test locations and include several scenes from each.

The local quality mask reduces problems from clouds and missing data, although its classifications can be wrong. Restricting the reference set to complete buildings gives a consistent boundary rule but excludes some visible objects from evaluation. Map edits and differences between map and image dates add uncertainty. Saving the inputs makes these decisions traceable, while OSM coverage and accuracy remain limitations of the reference data.

Each corrected run saves the crop, its mask, eligible footprints, predictions, raw model output and a settings manifest. It also records a source snapshot, dependency versions and file checksums. Replaying these files avoids changes from live services. Network failures are recorded as failed attempts and are not converted into zero detection scores.

7. Conclusion and Future Work

I built an application that combines a colour rule, OWL ViT and a hybrid to investigate red roof detection. Across the seven crops, the colour candidates had limited overlap with OSM buildings. Neither OWL ViT nor the hybrid produced a matched prediction. Looking at the intermediate outputs helped explain the empty results: some candidates fell below the confidence threshold, while others were too large to pass the size filter. The bird photograph and two clearer overhead images produced successful localisations, showing that empty satellite outputs were not the model's only behaviour. The results describe the imagery, prompts and filters tested here. Evaluating roof colour itself still requires labelled roofs.

Before collecting more results, I would choose a single evaluation target. Continuing with red roofs would require clearer imagery and independently checked colour labels, with a rule for ambiguous examples. A broader building detection study would need prompts and references aimed at buildings generally. In either case, settings should be chosen on development scenes and

evaluated on fresh scenes. The application and archived checks provide a practical base for that next experiment.

Author Contribution

I proposed the project and am the sole author of this paper. I take responsibility for the submitted work and its interpretation. The internship provided the setting and study time for the project; mentioning BİTES and the NCIA request does not imply that either organisation reviewed or endorsed the findings.

Code and Data Availability

The project repository is listed at github.com/erenesinler23/geoai-feature-detection. The exact experiment inputs, predictions, settings, source snapshots and file checksums are retained in the run archives. These archives allow the matching results to be reproduced without querying live services. The seven experiment archives are retained locally; the live examples in Appendix A are identified separately by their saved run folders. These archives are not included in the public source repository. A separate diagnostic archive is also retained locally. It contains the raw outputs, prompt settings, scripts, external control inputs, matching results and checksums from 21 September 2026. The scripts retain the local project paths used for the checks and require adaptation for use as a standalone installation package.

Appendix A The Interactive Web Application and Running the Code

The five screenshots below show the updated application on 20 September 2026. They cover the homepage, a London results page, a Prague comparison, the enlarged London colour output and a cached Dubrovnik result. Their displayed metrics were checked against the corresponding saved results. They document the application in use. Assessing geographic generalisation would require an independent set of locations.

A.1 Search Interface and Locations

The Flask application accepts a place name and geocodes it with Nominatim. It analyses a crop around the returned point, rather than the full boundary of a city or region. Searches are not restricted to the seven experiment locations. A live run still depends on usable imagery, mapped buildings and responses from the external services.

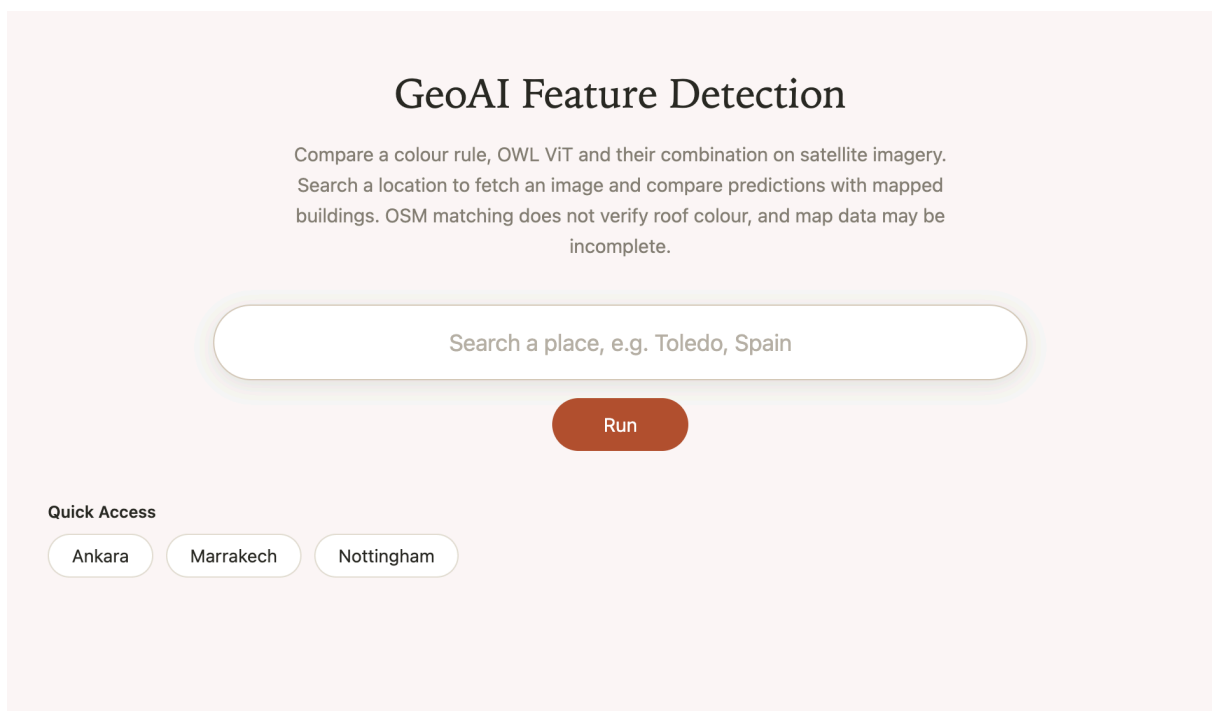


Figure 7. Updated homepage. No location is selected. Quick Access offers Ankara, Türkiye; Marrakech, Morocco; and Nottingham, United Kingdom.

A.2 London Results

Figure 8 shows London, United Kingdom. The colour rule retained 1,111 predictions and matched 83 of 8,863 eligible buildings. Precision was 7.47%, recall 0.94% and F1 1.66%. OWL ViT produced one candidate above the confidence threshold, but it failed the size filter. No model or hybrid predictions remained. The screenshot ends at their metric labels.

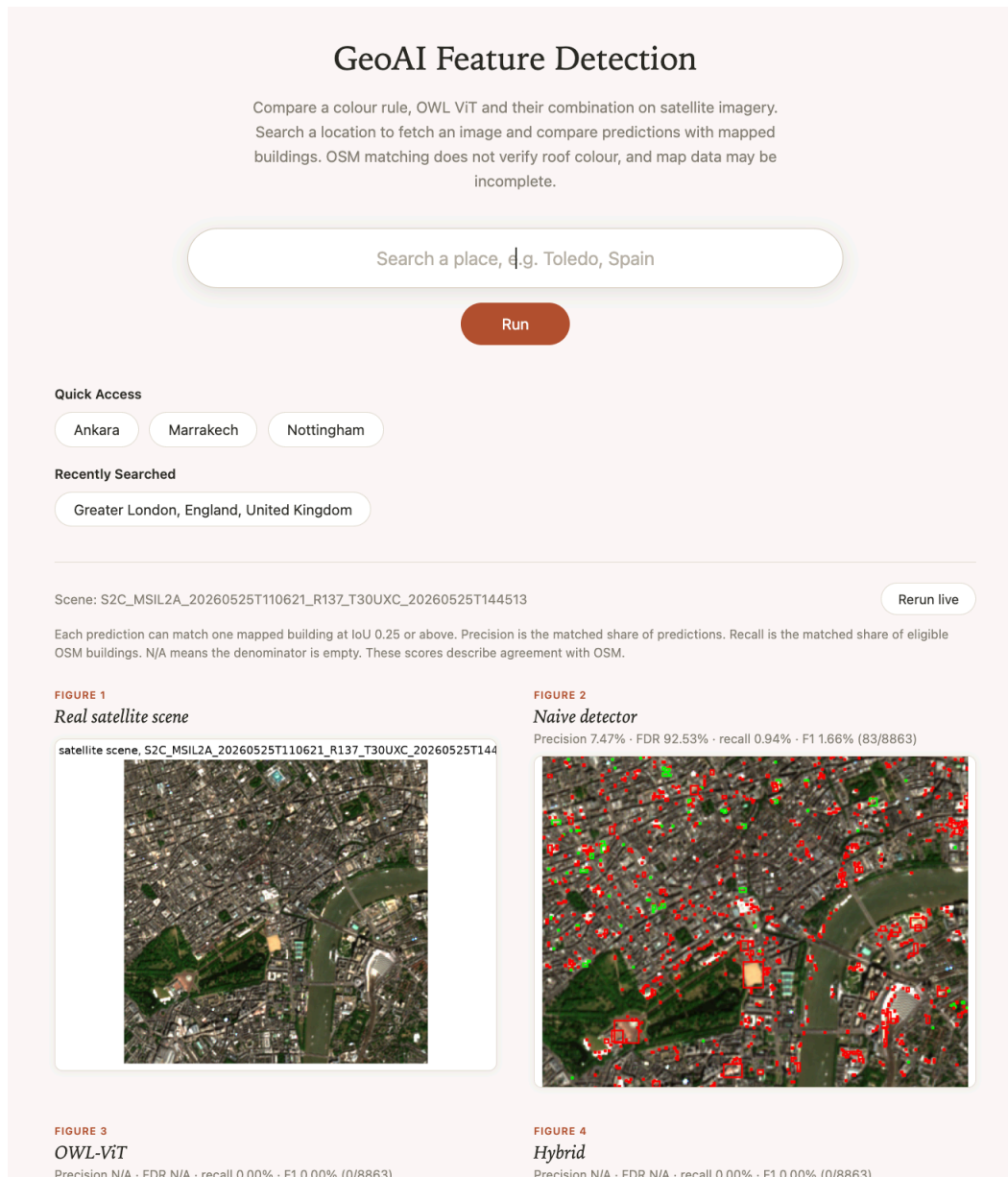


Figure 8. London results page. Sentinel 2 scene acquired on 25 May 2026. These are live application results, separate from the fixed London experiment in Tables 1 to 3.

A.3 Prague Comparison



Figure 9. Prague, Czechia, shown as Praha, Česko in the interface. Sentinel 2 scene acquired on 14 August 2026. All three detector panels are visible. The colour rule matched 332 of 5,738 eligible buildings.

A.4 Enlarged London Image

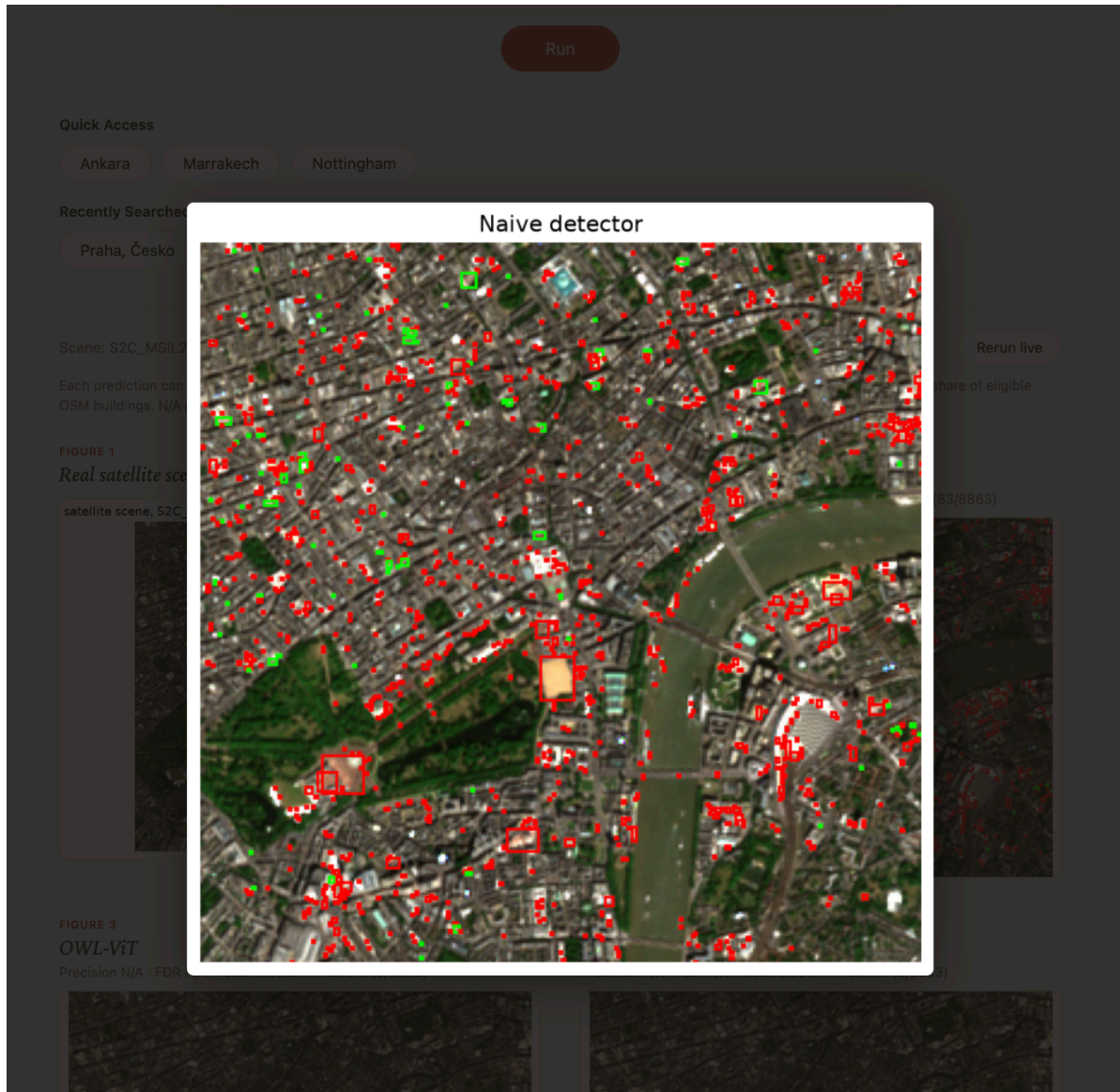


Figure 10. Enlarged colour detector output for London, United Kingdom, from the same live result as Figure 8. Green boxes are matched predictions; red boxes are unmatched predictions under the stated IoU rule.

Clicking an image opens the enlargement view. The larger view makes candidate locations easier to inspect while retaining the original image detail and leaving roof colour unverified. The visible Prague entry in the recent search list is a navigation option; the enlarged result itself is London.

A.5 Cached Dubrovnik Result

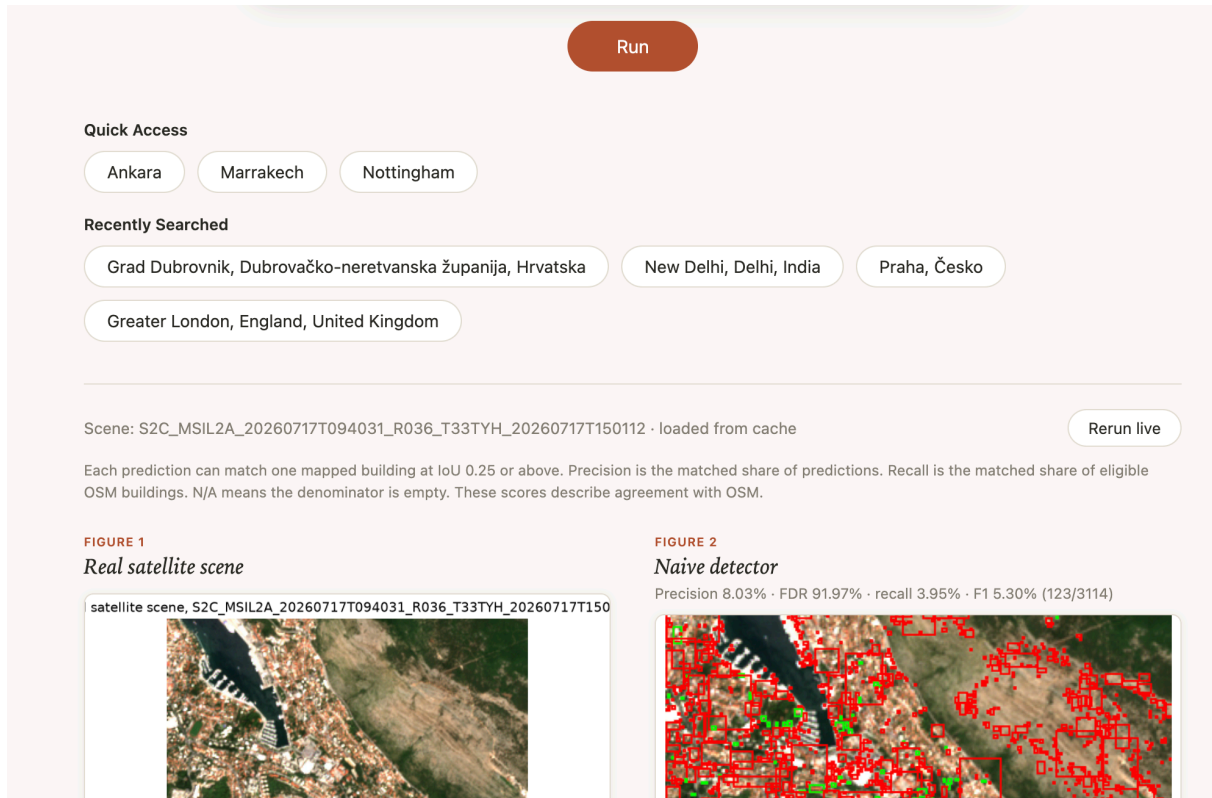


Figure 11. Cached result for Dubrovnik, Croatia, shown as Grad Dubrovnik in the recent search list. Sentinel 2 scene acquired on 17 July 2026. The label “loaded from cache” confirms reuse of the stored result.

The Dubrovnik colour rule retained 1,531 predictions and matched 123 of 3,114 eligible buildings. Precision was 8.03%, recall 3.95% and F1 5.30%. Dubrovnik is outside the seven location experiment and is included here as an application example. The recent search list also contains New Delhi, Prague and London; those labels do not describe the active image.

Recent results are stored in memory while the application is running. A repeated search reuses the result and its saved images. Rerun live bypasses that cache and starts a new run. Restarting the application clears the in memory cache. Each completed run still has a separate folder on disk. Live searches run one at a time while cached pages and saved images can be served by other requests.

A.6 Records for the Webpage Examples

The London example uses `outputs/london/20260920T181634_cf43751e`. The Prague example uses `outputs/prague/20260920T181857_c13cfccf`. The Dubrovnik example uses `outputs/dubrovnik/20260920T182310_d2f37494`. Each folder contains the scene identifier, image, eligible footprints, predictions, results and settings. Their saved metric counts match the screenshots.

All three live examples use colour thresholds of 120, 25 and 25, model confidence 0.01 and matching IoU 0.25. The live searches use geocoded points, whereas `benchmark.json` fixes the experiment centres. Even when the same satellite scene is selected, a different crop or map snapshot can change the retained predictions and reference count. The London and Prague differences therefore are not evidence of a change in model quality.

A.7 Requirements and Setup

The verified environment uses Python 3.14.7. The requirements file pins the package versions used. Live searches need internet access. Saved matching results can be replayed offline. The following commands create and activate a local environment on macOS or Linux.

```
python3 -m venv .venv
source .venv/bin/activate
python -m pip install -r requirements.txt
```

In VS Code, select the Python interpreter inside this environment so that the editor and terminal use the same installed libraries. The model loads once per application process. It selects Apple MPS when available, then CUDA, otherwise CPU. The terminal reports the actual model device when loading finishes. GPU speed was not benchmarked in this study.

A.8 Configuration and Reproduction

Default live settings are in `config.py`. The fixed experiment locations and shared thresholds are in `benchmark.json`. The Nottingham Quick Access preset retains its earlier colour thresholds of 140, 40 and 40 and model confidence 0.02. It therefore differs from the shared experiment settings of 120, 25 and 25 and confidence 0.01. Use the batch configuration when reproducing the paper.

```
python run_benchmark.py
python replay_evaluation.py outputs/verified/marrakech
```

The first command fetches a fresh batch, which can differ if external data changes. The second repeats the matching calculation on saved predictions, including sensitivity to the IoU threshold. The `rerun_saved.py` script can also repeat detection on archived imagery. Before running it, set `OWL_MODEL_REVISION` to the revision in the saved manifest and check that the model settings match the archived run. The synthetic demo is a software check and is not evidence of performance on satellite data.

```
python demo.py
```

A.9 Running the Web Application

Activate the environment and start the application. The default browser address is `http://127.0.0.1:5055`. This local address uses port 5055 to avoid the port 5000 conflict encountered during development.

```
source .venv/bin/activate
python app.py
```

The source archive includes project code without copying the installed libraries. Plotting reuses a coordinate converter across the detected boxes. These changes reduce unnecessary work while keeping the detection settings and matching rules unchanged. The terminal prints stage timings for each completed live run. No general response time or acceleration claim is made here, since service delays and hardware vary.

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